

Hedyotis diffusa alleviate aflatoxin B1-induced liver injury in ducks by mediating Nrf2 signaling pathway

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ABSTRACT

Aflatoxin B1 (AFB1), the most harmful aflatoxins, is a frequent contamination in feed and food items, raising global concerns in animal production and human public health. Also, AFB1 induces oxidative stress, cytotoxicity, mutations, and DNA lesions through its metabolic transformation into aflatoxin B1-8,9-epoxide (AFBO) by cytochrome P450 (CYP450). *Hedyotis diffusa* (HD) is a traditional Chinese herbal medicine known for its multiple pharmacological activities, including antioxidant, anti-inflammatory, and immunomodulatory. Yet, the influence of HD on AFB1-induced liver injury in ducks is still unknown. Here, we investigated whether HD positively affects AFB1-induced liver injury in ducks. Results revealed that I) AFB1 caused significant changes in serum biochemical indices and decreased growth performance of ducks (such as ALT, AST, ALP, TP, ALB, final body weight, and body weight gain), whereas HD supplementation at 200 mg/kg mitigated these alterations. II) HD alleviated hepatic histopathological changes and liver index induced by AFB1 in ducks. III) HD significantly attenuated AFB1-induced oxidative stress, as measured by increased antioxidant enzyme activities such as SOD, GPx, and T-AOC and decreased MDA levels. Furthermore, HD reduced the level of AFB1-DNA adduct in duck liver. IV) HD significantly promoted the transcriptional expression of NF-E2-related nuclear factor 2 (Nrf2) and associated genes, including heme oxygenase 1 (HO-1), NAD(P)H dehydrogenase quinone 1 (NQO1), glutamate-cysteine ligase catalytic (GCLC). In conclusion, these results demonstrated that HD could activate the Nrf2 pathway in ducks to reduce the hepatotoxicity driven by AFB1. This finding also provides theoretical and data support for a deeper understanding of the toxic mechanisms of AFB1 and its prevention.

1. Introduction

Aflatoxins are a class of mycotoxins generated by the genus *Aspergillus*, particularly *A. flavus*, and are secondary fungal metabolites (Rushing and Selim, 2019). Following a "Turkey X illness" outbreak in England, aflatoxins were initially identified (Blount, 1961). Since its appearance, aflatoxins have made direct or indirect losses that exceed billions of dollars each year, according to the statistical data from World Food and Agriculture Organization (FAO) (Cao et al., 2021). The aflatoxin class has four compounds, including aflatoxin B1, B2, G1, and G2 (Asao et al., 1963), of which AFB1 is the most toxic and harmful. It has been demonstrated that exposure to AFB1 can lead to malnutrition, growth impairment, immunosuppression and genotoxicity, more severely, may result in carcinogenicity and death in animals (Rushing and Selim, 2019). According to previous studies, the metabolic transformation of AFB1 represents the primary cause of its toxicity.

Mechanistically, AFB1 is ultimately transformed into AFBO in the liver by the cytochrome P450 (CYP450) enzyme system, AFBO then forms adducts with proteins and DNA that cause cytotoxicity, mutations, and DNA damage (Rushing and Selim, 2019). Additionally, the metabolic transformation of AFB1 can also induce oxidative stress, and generate a great proportion of reactive oxygen species (ROS), ROS finally may result in the oxidation of DNA, proteins, and lipids (Frijhoff et al., 2015). AFB1 is also listed by the International Agency for Research on Cancer as a Class I carcinogen (Pu et al., 2021). More dangerously, AFB1 may endanger human health by consuming foods obtained from animals (such as animal tissues, milk, and eggs). AFB1 has thus sparked worries about animal productivity and public health worldwide. Until now, despite many researchers' efforts, there are still no efficient preventive measures to lessen the toxicity of AFB1. Thus, finding a safe and effective strategy to protect humans and animals undergoing the toxicity of AFB1 is urgently needed.

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Recently, plant extracts have been utilized to prevent or cure many illnesses caused by toxins, viruses, and bacteria, due to the multiple pharmacological activities and fewer side effects (Garber et al., 2021; Qian et al., 2021; Oulahal and Degraeve, 2022). HD is a natural medicinal plant, widely abundant in tropical and subtropical regions (Li et al., 2013), and it is also a traditional Chinese herbal medicine with thousands of years of history. HD is composed of alkaloids, anthraquinones, lignans, triterpenes, flavonoids, coumarins as well as iridoids and it has been confirmed that HD has multiple pharmacological activities, such as anti-inflammatory, antioxidant, immunomodulatory, anticancer and antitumor, etc. (Ahmad et al., 2005; Ye et al., 2015; Qian et al., 2021). Since its various biological properties, HD has been widely used for treating inflammation-related diseases including hepatitis, bronchitis, urethritis, appendicitis and has been suggested as a possible medicine for tumors and cancers (Chen et al., 2016; Qian et al., 2021).

Interestingly, some studies have proved that HD can prevent rat from genotoxicity and carcinogenicity induced by AFB1 via inhibiting CYP450 enzyme system metabolizing AFB1 into the highly reactive AFBO (Wong et al., 1993). However, its capacity to mitigate AFB1-induced oxidative stress is still elusive. Nrf2 is a nuclear transcription factor that plays a crucial role in antioxidation and detoxification via promoting the transcriptional expression of associated genes such as glutathione S-transferase (GST) family, HO-1, NQO1 and GCLC (Benkerroum, 2020). Therefore, Nrf2 factor is considered a potential target to reduce the toxicity of AFB1. So far, few studies focus on HD's potential to influence antioxidant capacity by activating Nrf2 pathway in birds. As demonstrated in previous studies, ducks are sufficiently susceptible to AFB1, which could be used widely in a protective efficacy bioassay for AFB1 (Patterson, 1973). Considering the susceptibility of ducks to AFB1, Pekin-ducks were used to perform this trail. In general, the current study deeply evaluated whether HD has a protective impact on liver damage caused by AFB1 in ducks and its underlying mechanism.

2. Materials and methods

2.1. Major Chemicals and Reagents

FERMENTEK Ltd. (Jerusalem, Israel) provided standard AFB1 (purity 99.5%). The extract of HD plant was purchased from Dostov Biotechnology Co., Ltd. (Nanjing, China). A 1000 g dried entire plant of HD was extracted with ethanol for two hours and repeated three times. After, the extraction fluid was concentrated and dried under a vacuum to give 100 g of HD extract. Methanol, DMSO (dimethyl sulfoxide), and Sinopharm Chemical Reagent Co., Ltd. (Shanghai, China) supplied other chemical reagents.

2.2. Animals, Treatments, and Sample Collection

The trial was carried out following the protocol authorized by the Institutional Animal Care Committee of Huazhong Agricultural University, China. (Approval number: SYXK(E)2020-0084). 120 male Pekin ducks at one-day-old were purchased from Yongsheng Duck Industry Co., Ltd. (Wuhan, China). With six replicates of five ducks each, they were randomly divided into four experimental groups. The light/dark cycle was set to 12 h, and the ducks were given a standard commercial basic diet and unlimited access to fresh water. Ducks were fed one of four different diets after four days of adaptive feeding: basic diet (Control); basic diet plus 50 µg/kg AFB1 (AFB1); basic diet plus 50 µg/kg AFB1 and 200 mg/kg HD (AFB1 +HD); or basic diet plus 200 mg/kg HD (HD). The preparation of AFB1-contaminated diet was based on previously reported. Briefly, 10 mg AFB1 was dissolved in 10 ml methanol and evenly sprayed into 2 kg of diet to obtain an AFB1-contaminated diet with 5 mg/kg AFB1. It was then doubled dilution AFB1 concentration to 50 µg/kg. The experiment duration and the dosage of AFB1 were chosen based on previous studies that indicated exposure to 40–110 µg/kg AFB1 can significantly impair liver function and affect the

growth performance of Pekin-ducks (Han et al., 2008; Chen et al., 2014). The ducks were fed for 14 days, and weight every 4 days. Six ducks were euthanized at the end of the experiment to obtain blood and liver samples. Ducks were chosen at random. (1 duck from each replication). Blood sample was centrifuged at 1000 rpm/min at 4 °C for 10 min to get serum and kept at – 20 °C until use. Livers were washed by saline, dried with absorbent paper and weighed, then snap-frozen in liquid nitrogen, and kept at – 80 °C until use.

2.3. Serum biochemical parameters analysis

The serum was gently thawed at 4 °C before measurement. The serum biochemical parameters like aspartate amino transferase (AST), alanine amino transferase (ALT), alkaline phosphatase (ALP), gamma-glutamyl transpeptidase (GGT), albumin (ALB) and total protein (TP) were determined using an automatic biochemistry analyzer (BS-240VET, Mindray Bio-Medical Co., Ltd. ShenZhen, China).

2.4. Histologic analysis

The liver tissues were embedded in paraffin after being fixed in 10% neutral formalin, dried in graded ethanol solutions, then cut at 5 µm and stained with hematoxylin and eosin (HE) to diagnose histological alterations (Wang et al., 2021).

2.5. Liver antioxidant abilities analysis

Malondialdehyde (MDA) concentrations and the activity of SOD, glutathione peroxidase (GPx), and T-AOC were measured with assay kits (A001, A005, A015–2, and A003) obtained from the Nanjing Jiancheng Bioengineering Institute (Nanjing, China). Protein concentrations were measured using the bicinchoninic acid method with BCA assay kit (P0010s) manufactured by Beyotime Biotechnology (Shanghai, China).

2.6. Measurement of AFB1 in Diet and AFB1-DNA Adducts in Liver

AFB1's concentrations in diet were measured using ELISA kit (10002122) purchased from Romer Labs (Getzersdorf, Austria). The hepatic AFB1-DNA adduct concentrations were detected by ELISA kit (MM-9520601) manufactured by Meimian Industrial Co., Ltd (Jiangsu, China).

2.7. Related gene levels analysis

Total RNA was collected from 100 mg liver samples (6 samples per group) using TRIzol reagent (AG21101, Accurate Biotechnology Co., Ltd. Hunan, China) in accordance with the manufacturer's instructions. Using Thermo Scientific NanoDrop 2000 (Madison, USA), the extracted RNA's concentrations, and quality were assessed. The Evo M-MLV Mix Kit was used for reverse transcription in accordance with the manufacturer's guidelines (AG11728, Accurate Biotechnology Co., Ltd. Hunan, China). After that, SYBR Green Pro Taq (AG11701, Accurate Biotechnology Co., Ltd. Hunan, China) was used to perform RT-qPCR with Step One-Plus Real-time PCR System (Applied Biosystems, Carlsbad, USA). The relative abundance was adjusted to the Control group and quantified using the $2^{-\Delta\Delta C_t}$ methodology with β -actin as a reference gene. Primers for β -actin gene and Nrf2 signaling-related genes were provided and synthesized by Sangon Biotech Co., Ltd. (Shanghai, China). Specific primer sequences are displayed in [Supplementary Table 1](#).

2.8. Western Blotting

BCA Assay Kit was used to measure the protein concentrations. SDS-polyacrylamide gel electrophoresis was used to separate the proteins, then proteins were transferred to nitrocellulose membranes. Blocking buffer was applied to blocked membranes for 1 h at room temperature

(tris buffered saline (TBS) with 0.1% Tween 20% and 5% nonfat dry milk). Incubate membrane and primary antibody in 10 ml blocking buffer with gentle agitation overnight at 4 °C (The manufacturers, catalogue numbers and dilution ratio of primary antibodies are displayed in [Supplementary Table 2](#)). Following that, the membranes were incubated with secondary antibodies for one hour at room temperature (RT). (Goat anti-rabbit and goat anti-mouse, detailed information was shown in [Supplementary Table 2](#)) After further washing by TBS PLUS Tween 20, the membranes were incubated with commercial chemoluminescence reagents (Vazyme Biotech Co., Ltd. Nanjing, China) and exposed to X-ray to get images. Image J (National Institutes of Health, Maryland, USA) software was used to calculate band intensities. Protein levels were respectively normalized to β -actin.

2.9. Statistical analysis

Prism software was used to analyze all data by Student's t-test or one-way ANOVA (Version 8.4.2, GraphPad Software, Inc.). The results were presented as means $\pm$ standard deviation (SD). $P < 0.05$ was regarded as statistically significant.

3. Results

3.1. AFB1 concentration, growth performance and serum biochemistry

The overall AFB1 concentration and growth performance results are shown in [Table 1](#). Before the trial, the residues of AFB1 were measured using the AFB1-ELISA kit to make sure different treatment ducks get exposed to different AFB1. As expected, there were significant differences in the basal diet AFB1 concentrations among different groups ($P < 0.05$), specifically, 10 mg/kg in control and HD groups while 60 mg/kg in other groups. In terms of growth performance, there were no differences ($P < 0.05$) in the initial body weight within the four groups. However, AFB1 reduced the duck's ultimate body weight and weight gain ($P < 0.05$) after 14 days of experimental treatments. HD supplementation at 200 mg/kg substantially attenuated ($P < 0.05$) these AFB1-induced effects. In [Table 2](#), the serum biochemistry result is displayed. The AFB1 group had substantially higher ($P < 0.05$) serum levels of AST, ALT, ALP, and GGT. At the same time, AFB1 sharply reduced ($P < 0.05$) the levels of ALB and TP. Interestingly, these biochemistry changes were also alleviated ($P < 0.05$) by HD. By the way, HD supplementation at 200 mg/kg had no statistical effects on ducks' growth performance and serum biochemistry compared with the control group ([Fig. 1](#)).

3.2. Liver Index and Histology

AFB1 significantly increased ($P < 0.05$) the hepatic index, while HD supplementation could significantly mitigate the phenomenon ([Fig. 2](#)). Meanwhile, we used HE staining to observe differences in liver structure

Table 1

Effect of AFB1 and HD on growth performances of ducks¹.

	Control	AFB1	AFB1 +HD	HD
Initial body weight, g	63.83 $\pm$ 3.31	64.17 $\pm$ 3.06	63.48 $\pm$ 1.00	64.33 $\pm$ 1.21
Final body weight, g	552.50 $\pm$ 76.66 ^a	424.17 $\pm$ 32.47 ^b	526.17 $\pm$ 26.46 ^{ac}	540.33 $\pm$ 44.21 ^{ac}
Body weight gain, g	488.67 $\pm$ 73.47 ^b	360.00 $\pm$ 29.65 ^a	462.68 $\pm$ 25.50 ^{bc}	476.00 $\pm$ 43.12 ^{bc}
residues of AFB1, μ g/kg	13.12 $\pm$ 1.51 ^a	60.75 $\pm$ 18.76 ^b		13.12 $\pm$ 1.51 ^a

¹Note: All values were expressed as mean $\pm$ standard deviation (n = 6). Different letters indicated statistically significant differences ($P < 0.05$). Control, basal diet; AFB1, basal diet supplement with 50 μ g/kg AFB1; AFB1 +HD, basal diet supplement with 50 μ g/kg AFB1 plus 200 mg/kg HD; HD, basal diet plus 200 mg/kg HD.

Table 2

Effect of AFB1 and HD on serum biochemistry index of ducks¹.

	Control	AFB1	AFB1 +HD	HD
AST activity, U/L	15.18 $\pm$ 2.39 ^a	23.65 $\pm$ 3.07 ^b	18.13 $\pm$ 2.75 ^{ac}	18.10 $\pm$ 1.83 ^c
ALT activity, U/L	40.78 $\pm$ 3.65 ^a	49.88 $\pm$ 1.66 ^b	44.30 $\pm$ 2.64 ^a	42.78 $\pm$ 2.79 ^a
ALP activity, U/L	2044.87 $\pm$ 566.16 ^a	2869.73 $\pm$ 189.04 ^b	2370.75 $\pm$ 335.90 ^a	2139.18 $\pm$ 355.85 ^a
GGT activity, U/L	3.25 $\pm$ 0.78 ^a	4.83 $\pm$ 0.31 ^b	3.25 $\pm$ 0.59 ^a	3.27 $\pm$ 0.42 ^a
ALB, g/L	16.65 $\pm$ 1.32 ^a	11.62 $\pm$ 0.67 ^b	13.78 $\pm$ 0.67 ^c	15.92 $\pm$ 0.76 ^{ad}
TP, g/L	45.55 $\pm$ 3.78 ^a	34.58 $\pm$ 1.15 ^b	40.20 $\pm$ 1.53 ^c	41.60 $\pm$ 2.28 ^{ac}

¹Note: All values were expressed as mean $\pm$ standard deviation (n = 6). Different letters indicated statistically significant differences ($P < 0.05$). AST, Aspartate amino transferase; ALT, Alanine amino transferase; ALP, Alkaline phosphatase; GGT, Gamma glutamyl transpeptidase; ALB, Albumin; TP, Total protein. Control, basal diet; AFB1, basal diet supplement with 50 μ g/kg AFB1; AFB1 +HD, basal diet supplement with 50 μ g/kg AFB1 plus 200 mg/kg HD; HD, basal diet plus 200 mg/kg HD.

among various groups to evaluate the impact of AFB1 on the liver. According to the histopathology results, AFB1 exposure caused severe liver damage such as vacuolar degeneration, inflammatory cell infiltration, cell necrosis, and swelling, which were exactly the reasons for causing the increase in the liver index by AFB1 ([Fig. 2 A](#)). Conversely, HD supplementation reduced the hepatic damage caused by AFB1.

3.3. Hepatic Antioxidant Capacity and AFB1-DNA Adduct Concentration

The levels of MDA and the activity of antioxidant enzymes (T-AOC, SOD, GPx) in duck livers were measured to investigate the effects of AFB1 on oxidative damage. [Fig. 3](#) shows that, as compared to the control group, AFB1 decreased ($P < 0.05$) the activities of T-AOC, SOD, and GPx, but increased ($P < 0.05$) the level of MDA in duck livers. Notably, AFB1-induced alterations were significantly attenuated by HD supplementation. In addition, HD supplementation individually at 200 mg/kg had no statistical impact on T-AOC, SOD, and MDA concentrations compared to the control group. However, it reduced ($P < 0.05$) GPx enzyme activity. Another notable difference was that dietary supplementation with HD decreased ($P < 0.05$) the level of AFB1-DNA adducts, whereas AFB1 improved ($P < 0.05$) the level of AFB1-DNA adducts in duck livers, as demonstrated in [Fig. 2B](#).

3.4. The HD alleviated AFB1-induced oxidative damage via promoting Nrf2 pathway

Due to the significant differences in antioxidant capacity among different groups, we detected *Nrf2*, the key regulator during oxidative stress ([Hayes et al., 2009](#)), and its related genes expression. As shown in [Fig. 4](#), AFB1 decreased ($P < 0.05$) the mRNA levels of *Nrf2*, *GCLC*, and *HO-1* while increasing ($P < 0.05$) the mRNA level of *NQO1*. In contrast to AFB1 group, HD increased ($P < 0.05$) the mRNA levels of *Nrf2*, *HO-1* and *GCLC*. Moreover, Western blot results revealed that AFB1 and HD treatment statistically affected the protein levels of Nrf2 and associated genes in duck livers ([Fig. 5](#)). Specifically, AFB1 exposure downregulated ($P < 0.05$) the protein levels of Nrf2, GCLC, HO-1 and NQO1. However, HD treatment significantly improved ($P < 0.05$) these protein levels. Notably, exposure to AFB1 resulted in an upward trend in the mRNA level of *GCLC* gene, but it downregulated the protein level of GCLC gene in duck livers.

4. Discussion

AFB1, a prevalent and harmful mycotoxin, typically contaminates

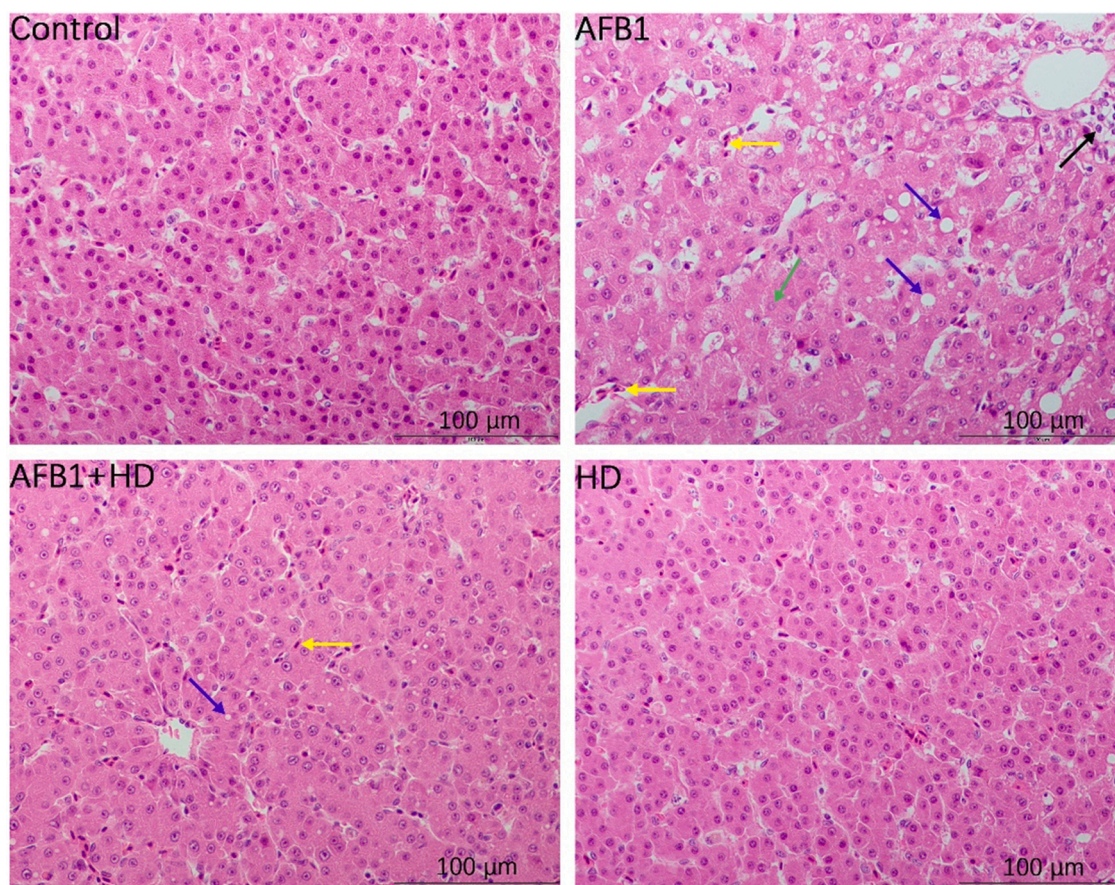


Fig. 1. Effect of AFB1 and HD on histopathology in the livers of ducks. (200 × magnification) Note: black arrows represent inflammatory cell infiltration; purple arrows represent vacuolar degeneration; yellow arrows represent blood cell oozing; green arrows represent nuclear pyknosis. Control, basal diet; AFB1, basal diet supplement with 50 μg/kg AFB1; AFB1 +HD, basal diet supplement with 50 μg/kg AFB1 plus 200 mg/kg HD; HD, basal diet plus 200 mg/kg HD.

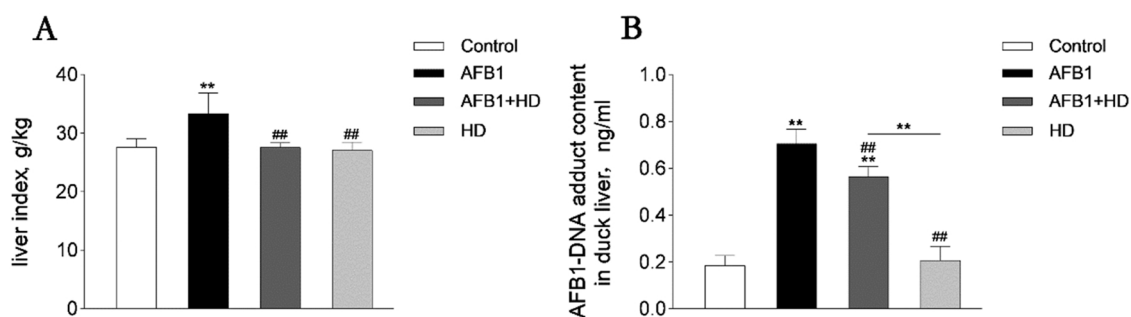


Fig. 2. Liver index and residues of AFB1-DNA adduct in the livers of ducks. (A) Effect of AFB1 and HD on the liver index of ducks. (B) Residues of AFB1-DNA adducts in the liver of ducks. All data were expressed as mean ± standard deviation (n = 6). * represent statistical difference compared to control group; # represent statistical difference compared to AFB1 group. * 0.01 < P < 0.05, ** P < 0.01; # 0.01 < P < 0.05, ## P < 0.01.

feedstuffs and cereal grains, endangering human and animal health (Barati et al., 2018; Rushing and Selim, 2019). Although we have understood the mechanisms of toxicity of AFB1 to some extent, there are still no effective preventive measures (Benkerroum, 2020). This study examined the effect of HD on hepatotoxicity caused by AFB1 in ducks using factors such as growth performance, serum biochemistry, liver histology, and hepatic antioxidant capacity. The results indicated that HD could significantly reduce liver damage in ducks which could be utilized to prevent aflatoxicosis in ducks.

Birds exposed to AFB1 exhibited poor growth performance. It has been reported that for every 1 mg/kg increase in AFB1 in the diet, the weight gain rate of broilers decreases by 5% (Dersjant-Li et al., 2003). This poor growth performance may be attributed to AFB1's adverse

effects, such as altering the gastrointestinal tract's structure and microbial community, decreasing the activity of digestive enzymes and altering the intestinal immune status (Mughal et al., 2017). This damage will further reduce micronutrients, protein and glucose absorption (Benkerroum, 2020). In this study, the ducks in AFB1 treatment reduced final body weight and weight gain, which agrees with former studies (Chen et al., 2014). Notably, HD could improve this subpar growth performance.

An important marker of liver injury is alterations in serum biochemistry parameters, including ALT, AST, ALP, GGT, ALB, and TP. In regular clinical practice, ALT and AST are frequently used to assess liver damage (Hinson et al., 2010). It is generally assumed that increased serum ALT and AST levels are a direct consequence of release from

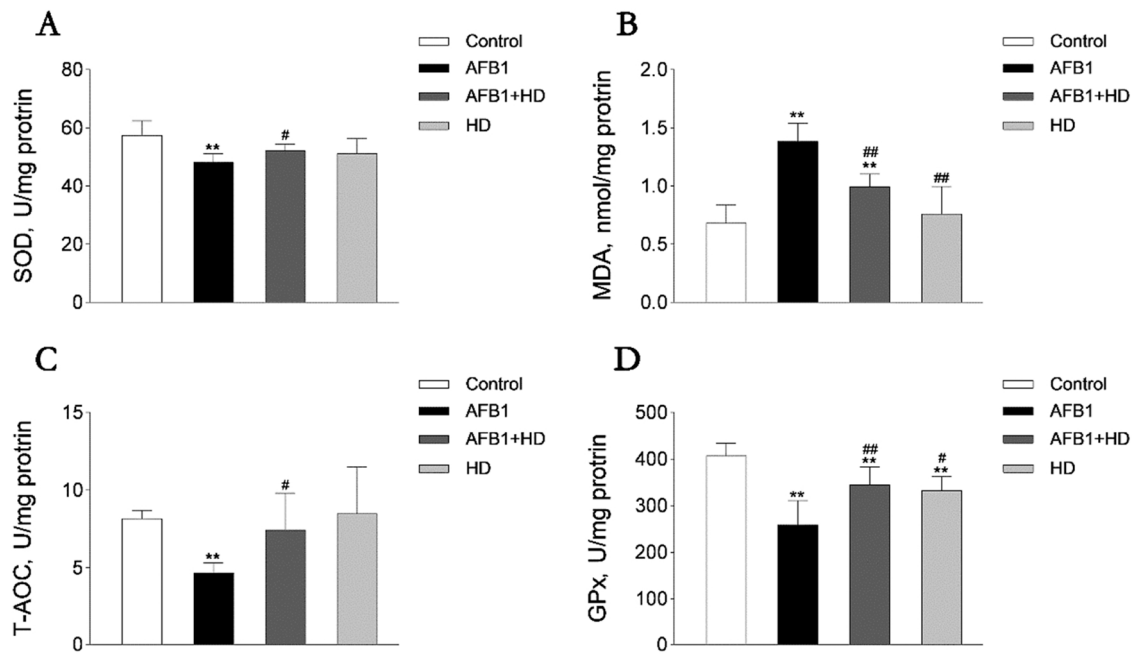


Fig. 3. Effect of AFB1 and HD on hepatic redox status of ducks. (A) SOD, Superoxide dismutase activity. (B) MDA, Concentration of malondialdehyde. (C) T-AOC, Total antioxidant capacity. (D) GPx, GSH-Px, Glutathione peroxidase activity. All data were expressed as mean $\pm$ standard deviation ($n = 6$). * represent statistical difference compared to control group; # represent statistical difference compared to AFB1 group. * $0.01 < P < 0.05$, ** $P < 0.01$; # $0.01 < P < 0.05$, ## $P < 0.01$.

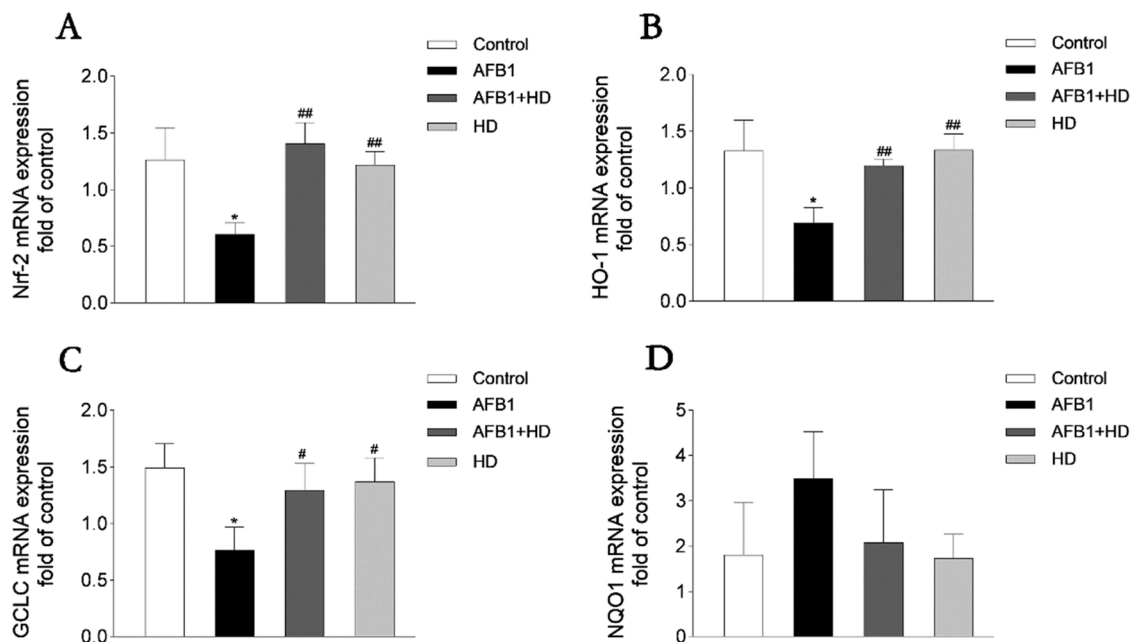


Fig. 4. Relative mRNA expression of *Nrf2*-related genes in the livers of ducks. (A) *Nrf2*, Nuclear factor-erythroid 2-related factor 2. (B) *HO-1*, Heme oxygenase 1. (C) *GCLC*, Glutamate-cysteine ligase catalytic subunit. (D) *NQO1*, NAD(P)H dehydrogenase quinone 1. All data were expressed as mean $\pm$ standard deviation ($n = 6$). * represent statistical difference compared to control group; # represent statistical difference compared to AFB1 group. * $0.01 < P < 0.05$, ** $P < 0.01$; # $0.01 < P < 0.05$, ## $P < 0.01$.

hepatocytes undergoing necrosis (Smith et al., 2020). ALP is initially produced in bone, then moved to liver by blood, and finally excreted through the biliary system. And GGT also comes from the hepatobiliary system primarily (Wang et al., 2021). The contents of TP and ALB in serum are representative indicators of hepatic secretory function. In this study, serum ALT, AST, ALP, and GGT activities increased significantly, revealing a release of these enzymes from hepatic damaged induced by AFB1. Moreover, the reduction of TP and ALB in serum showed strong

evidence of serious liver injury. In this study, dietary supplementation with HD at 200 mg/kg significantly mitigates serum biochemistry alterations. Interestingly it has been confirmed that HD could alleviate serum biochemistry parameters changes induced by AFB1 in mice and chickens as previous studies described, which remains in agreement with our studies (Dai et al., 2017; Zhao et al., 2021).

Generally, organ index and histopathological changes are used as indicators to evaluate organ injury (Ahmad et al., 2013). Taken liver as

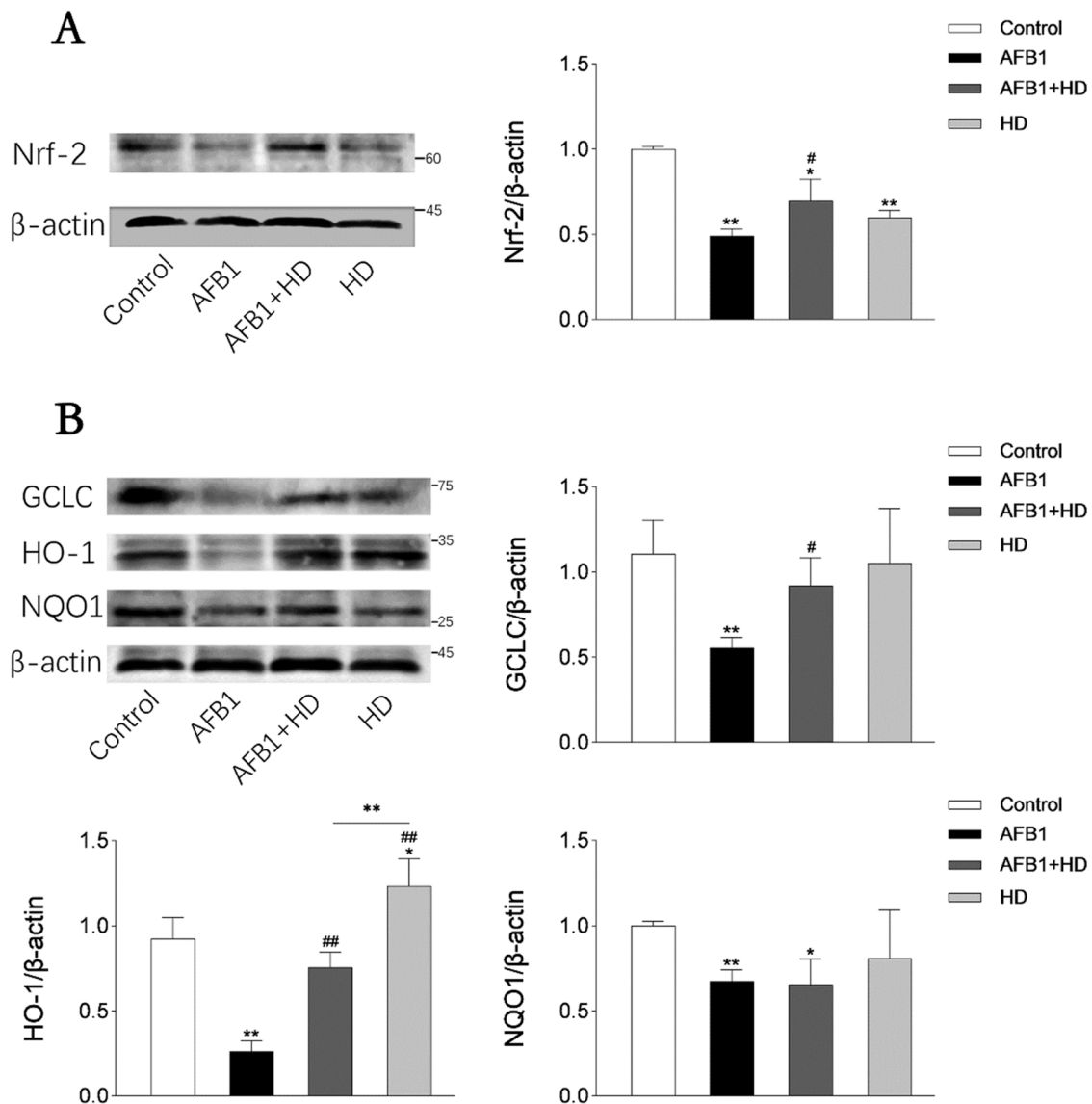


Fig. 5. The expression of Nrf2 and its related protein in the livers of ducks. (A) Nrf2, Nuclear factor-erythroid 2-related factor 2. (B) GCLC, Glutamate-cysteine ligase catalytic subunit; HO-1, Heme oxygenase 1; NQO1, NAD(P)H dehydrogenase quinone 1. Protein expression were respectively normalized to β-actin. * represent statistical difference compared to control group; # represent statistical difference compared to AFB1 group. * $0.01 < P < 0.05$, ** $P < 0.01$; # $0.01 < P < 0.05$, ## $P < 0.01$.

an example, injury induced by alcohol, bacterial infection or cancer is normally accompanied by hepatomegaly (Collier, 2019; Angulo and Lindor, 2002) And it has been confirmed that exposure to AFB1 can also result in hepatomegaly in mice (Chen et al., 2019). Moreover histopathological analysis in the previous study has shown that hepatic damage comes with vascular swelling and inflammatory cell infiltration, which may be one of the causes of hepatomegaly (Chen et al., 2019; Zhao et al., 2021). In the current study, AFB1 increased the liver index significantly. Along with the changes in inflammatory cell infiltration, steatosis, cell swelling, and necrosis found in the histopathological analysis, it suggested that AFB1 caused severe liver damage in ducks. Fortunately, previous studies have concluded that HD potentially prevents the liver from these histopathological changes in mice (Dai et al., 2017). That means HD could be useful in inhibiting the progression of hepatotoxicity. Interestingly, these histopathological changes were also reversed by HD in our trial, which is consistent with previous studies.

AFB1-DNA adduct is normally regarded as a bio-marker of DNA damage when humans and animals are exposed to AFB1 (Wang and Groopman, 1999; Engin, 2019; Li et al., 2019). The current study found

that HD significantly decreased the amount of AFB1-DNA in duck livers, which indicated that HD could contribute to the stability of DNA and detoxify mycotoxins in the liver. It is one of the main reasons why HD significantly reduced AFB1-induced liver damage in ducks.

Currently, there is growing evidence that oxidative stress contributes more to AFB1 toxicity (Benkerroum, 2020). Oxidative stress could enhance the generation of different forms of ROS, including oxygen free radicals like superoxide anion radical ($O_2^{\cdot-}$) and hydroxyl radical (OH), as well as nonradical oxidants like hydrogen peroxide (H_2O_2) and singlet oxygen (1O_2) (Zorov et al., 2014; Cao et al., 2021). Excessive ROS can result in protein, lipid and DNA oxidation, and generate large amounts of MDA, thereby causing cytotoxicity and genotoxicity (Frijhoff et al., 2015; Zhao et al., 2021). Additionally, oxidative stress was associated with a decline in oxidant-scavenging enzymes such as GPx and SOD (Sharma et al., 2007). The increased concentration of MDA and decreased enzyme activities of SOD, GPx and T-AOC caused by AFB1 revealed that ducks exposed to AFB1 in the current study suffered oxidative stress. It is interesting that HD could undo redox parameter modifications brought on by AFB1. Specifically, it could increase the

hepatic antioxidant enzyme activities of SOD, GPX, T-AOC compared with AFB1 groups. In summary, these results suggest that HD may reduce the oxidative stress caused by AFB1 in duck livers. These findings are consistent with former research that suggested HD could reduce oxidative damage brought on by various kinds of toxins (Ahmad et al., 2005; Chen et al., 2016; Gao et al., 2016).

The regulation of oxidative stress is a more complex process. For now, its mechanism is still poorly understood, yet a consensus among many researchers is that Nrf2/ARE signaling pathway hold a pivotal role in this process (Hayes et al., 2009; Dodson et al., 2019; Shaw et al., 2019). Nrf2 is a ubiquitously expressed transcription factor that modulates the levels of antioxidative and detoxifying enzymes, such as HO-1, NQO1, GCLC (Kundu and Surh, 2008). HO-1 are crucial enzymes that catalyze the rate-limiting step in heme catabolism. This progress will result in more bilirubin, a powerful antioxidant that protects cells from ROS-mediated damage, such as DNA damage (Shaw and Chattopadhyay, 2020). NQO1 are important detoxification enzymes that shield cells from oxidative injury caused by quinone. Furthermore, it can maintain cellular redox stability by keeping the reduced form of cellular antioxidants (Jeddi et al., 2017) and removing superoxide (Siegel et al., 2004). GCLC could promote the synthesis of GSH, and GSH normally acts as a cellular reductant. Specifically, the first step catalyzed by glutamate-cysteine ligase GCLC is the rate-limiting step in GSH synthesis, which means more GCLC could accelerate GSH synthesis (Zhang et al., 2018; Shaw and Chattopadhyay, 2020). Recent research suggests that Nrf2 is responsible for more than just cellular redox homeostasis. It also essential for DNA repair and proteostasis (Dodson et al., 2019). Upregulation of Nrf2 could prevent animal from oxidative stress and toxins by promoting the antioxidant capacity and detoxification ability. In the current study, AFB1 sharply downregulated the mRNA and protein abundances of Nrf2, HO-1, NQO1 and GCLC, which is consistent with previous research (Wang et al., 2021; Zhao et al., 2021). Notably, the abundance of these proteins was significantly increased by HD administration. In conclusion, by stimulating the expression of Nrf2 and associated genes, HD could considerably ameliorate the liver damage caused by AFB1 in ducks.

5. Conclusion

The supplementation with HD mitigated the adverse effect of AFB1 in ducks. Particularly, AFB1-induced liver damage was related to oxidative stress and its metabolic transformation, while HD could alleviate oxidative stress and improve the detoxification capacity of ducks via promoting the expression of Nrf2 and its related genes. Hence, HD could be an ideal therapeutic approach for preventing and treating AFB1 poisoning in ducks.

Ethics statement

The animal study was reviewed and approved by the Ethics Committee of the Huazhong Agricultural University.

CRediT authorship contribution statement

Pengpeng Wang: Conceptualization, Writing–original draft, Methodology, Data Curation. **Yingli Wang:** Validation, Writing–review & editing. **Tianyi Feng:** Visualization, Investigation, Formal Analysis. **Ziyin Yan:** Visualization, Investigation, Formal Analysis. **Di Zhu:** Visualization, Investigation, Formal Analysis. **Huixian Lin:** Visualization. **Dandan Deng:** Visualization. **Muhammad Fakhar-e-Alam Kulyar:** Writing–review & editing. **Mudassar Iqbal:** Writing–review & editing. **Yaoqin Shen:** Writing–review & editing, Resources, Project administration, Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ecoenv.2022.114339.

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